

Dataset link: <https://data.mendeley.com/datasets/x84p2g3k6z/1>

ResNet50

```
Classes: ['Phytophthora', 'Scab', 'Styler and Root']
Epoch 1/10 | Time: 41.7s
  Train -> Loss: 0.2821 Acc: 0.9013
  Val   -> Loss: 0.0987 Acc: 0.9727
-----
Epoch 2/10 | Time: 43.6s
  Train -> Loss: 0.1194 Acc: 0.9605
  Val   -> Loss: 0.3218 Acc: 0.9350
-----
Epoch 3/10 | Time: 44.9s
  Train -> Loss: 0.0882 Acc: 0.9690
  Val   -> Loss: 0.0547 Acc: 0.9853
-----
Epoch 4/10 | Time: 44.2s
  Train -> Loss: 0.0395 Acc: 0.9870
  Val   -> Loss: 0.2630 Acc: 0.9140
-----
Epoch 5/10 | Time: 44.4s
  Train -> Loss: 0.0394 Acc: 0.9906
  Val   -> Loss: 0.4386 Acc: 0.8721
-----
Epoch 6/10 | Time: 44.7s
  Train -> Loss: 0.1040 Acc: 0.9677
  Val   -> Loss: 0.0333 Acc: 0.9874
-----
Epoch 7/10 | Time: 44.7s
  Train -> Loss: 0.0415 Acc: 0.9847
  Val   -> Loss: 0.0976 Acc: 0.9686
-----
Epoch 8/10 | Time: 44.3s
  Train -> Loss: 0.0615 Acc: 0.9820
  Val   -> Loss: 0.0791 Acc: 0.9748
-----
Epoch 9/10 | Time: 44.6s
  Train -> Loss: 0.0418 Acc: 0.9897
  Val   -> Loss: 0.0855 Acc: 0.9665
-----
Epoch 10/10 | Time: 44.7s
  Train -> Loss: 0.0207 Acc: 0.9946
  Val   -> Loss: 0.0512 Acc: 0.9811
```

Final Test Accuracy: 98.12%
Total Test Images: 478

InceptionV3

```
Epoch 1/10 | 65.2s
  Train -> Loss: 0.5173 Acc: 0.8546
  Val   -> Loss: 0.1850 Acc: 0.9203
-----
Epoch 2/10 | 57.5s
  Train -> Loss: 0.2173 Acc: 0.9448
  Val   -> Loss: 0.2085 Acc: 0.9140
-----
Epoch 3/10 | 55.0s
  Train -> Loss: 0.1779 Acc: 0.9547
  Val   -> Loss: 0.2004 Acc: 0.9329
-----
Epoch 4/10 | 55.5s
  Train -> Loss: 0.1661 Acc: 0.9569
  Val   -> Loss: 0.1902 Acc: 0.9413
-----
Epoch 5/10 | 56.0s
  Train -> Loss: 0.1357 Acc: 0.9654
  Val   -> Loss: 0.1618 Acc: 0.9350
-----
Epoch 6/10 | 55.7s
  Train -> Loss: 0.0902 Acc: 0.9771
  Val   -> Loss: 0.0945 Acc: 0.9748
-----
Epoch 7/10 | 55.6s
  Train -> Loss: 0.0875 Acc: 0.9776
  Val   -> Loss: 0.0566 Acc: 0.9811
-----
Epoch 8/10 | 55.7s
  Train -> Loss: 0.0344 Acc: 0.9942
  Val   -> Loss: 0.0761 Acc: 0.9665
-----
Epoch 9/10 | 55.7s
  Train -> Loss: 0.1403 Acc: 0.9650
  Val   -> Loss: 0.0597 Acc: 0.9811
-----
Epoch 10/10 | 55.7s
  Train -> Loss: 0.0984 Acc: 0.9762
  Val   -> Loss: 0.1021 Acc: 0.9727
```

Final Test Accuracy: 97.07%

AlexNet

```
Epoch 1/10 | Time: 32.5s
  [Train] Loss: 1.2307, Acc: 0.3285
  [Val]   Loss: 1.0980, Acc: 0.3501
-----
Epoch 2/10 | Time: 21.8s
  [Train] Loss: 1.0996, Acc: 0.3411
  [Val]   Loss: 1.1110, Acc: 0.3501
-----
Epoch 3/10 | Time: 21.6s
  [Train] Loss: 1.0987, Acc: 0.3483
  [Val]   Loss: 1.1120, Acc: 0.3501
-----
Epoch 4/10 | Time: 21.9s
  [Train] Loss: 1.1001, Acc: 0.3559
  [Val]   Loss: 1.1005, Acc: 0.3501
-----
Epoch 5/10 | Time: 21.8s
  [Train] Loss: 1.0983, Acc: 0.3546
  [Val]   Loss: 1.1001, Acc: 0.3501
-----
Epoch 6/10 | Time: 21.7s
  [Train] Loss: 1.0967, Acc: 0.3658
  [Val]   Loss: 1.1003, Acc: 0.3187
-----
Epoch 7/10 | Time: 22.0s
  [Train] Loss: 1.0967, Acc: 0.3537
  [Val]   Loss: 1.1000, Acc: 0.3501
-----
Epoch 8/10 | Time: 22.4s
  [Train] Loss: 1.0976, Acc: 0.3636
  [Val]   Loss: 1.1020, Acc: 0.3187
-----
Epoch 9/10 | Time: 22.0s
  [Train] Loss: 1.0985, Acc: 0.3496
  [Val]   Loss: 1.1025, Acc: 0.3187
-----
Epoch 10/10 | Time: 22.1s
  [Train] Loss: 1.0969, Acc: 0.3577
  [Val]   Loss: 1.0981, Acc: 0.3501
```

Final Test Accuracy: 36.61%

CNN

```
Epoch 1/10 | Time: 35.4s
  Train -> Loss: 1.0184 Acc: 0.5664
  Val   -> Loss: 0.6251 Acc: 0.7358
-----
Epoch 2/10 | Time: 20.1s
  Train -> Loss: 0.5292 Acc: 0.7760
  Val   -> Loss: 0.4473 Acc: 0.8071
-----
Epoch 3/10 | Time: 20.6s
  Train -> Loss: 0.3556 Acc: 0.8618
  Val   -> Loss: 0.3182 Acc: 0.8742
-----
Epoch 4/10 | Time: 20.1s
  Train -> Loss: 0.2762 Acc: 0.8914
  Val   -> Loss: 0.2773 Acc: 0.8763
-----
Epoch 5/10 | Time: 20.2s
  Train -> Loss: 0.2032 Acc: 0.9219
  Val   -> Loss: 0.2469 Acc: 0.9140
-----
Epoch 6/10 | Time: 20.1s
  Train -> Loss: 0.1311 Acc: 0.9466
  Val   -> Loss: 0.1929 Acc: 0.9161
-----
Epoch 7/10 | Time: 20.1s
  Train -> Loss: 0.0926 Acc: 0.9695
  Val   -> Loss: 0.2127 Acc: 0.9371
-----
Epoch 8/10 | Time: 20.6s
  Train -> Loss: 0.0871 Acc: 0.9672
  Val   -> Loss: 0.1853 Acc: 0.9392
-----
Epoch 9/10 | Time: 20.3s
  Train -> Loss: 0.0515 Acc: 0.9852
  Val   -> Loss: 0.2796 Acc: 0.9161
-----
Epoch 10/10 | Time: 20.3s
  Train -> Loss: 0.0676 Acc: 0.9762
  Val   -> Loss: 0.3141 Acc: 0.9015
-----
```

Custom CNN Final Test Accuracy: 91.00%

VGG16

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	0.2787	0.8882	0.3446	0.8763
2	0.1130	0.9542	0.1263	0.9644
3	0.0693	0.9776	0.1601	0.9644
4	0.0488	0.9838	0.1021	0.9790
5	0.0393	0.9874	0.0739	0.9748
6	0.0322	0.9892	0.0714	0.9748
7	0.0061	0.9978	0.0729	0.9832
8	0.0587	0.9856	0.1056	0.9686
9	0.0528	0.9865	0.0560	0.9895
10	0.0561	0.9829	0.0300	0.9874

FINAL TEST ACCURACY: 98.54%

DEIT

Epoch	Tr. Loss	Tr. Acc	Val Loss	Val Acc
1	0.2273	0.9206	0.0863	0.9706
2	0.0275	0.9946	0.0432	0.9853
3	0.0035	1.0000	0.0325	0.9916
4	0.0012	1.0000	0.0320	0.9916
5	0.0007	1.0000	0.0326	0.9916
6	0.0005	1.0000	0.0330	0.9916
7	0.0004	1.0000	0.0335	0.9916
8	0.0003	1.0000	0.0342	0.9916
9	0.0002	1.0000	0.0347	0.9916
10	0.0002	1.0000	0.0352	0.9916

FINAL TEST ACCURACY: 98.74%

IMAGES details

Folder Type	Class Name	Count
Original	Phytophthora	114
Original	Scab	106
Original	Styler and Root	94
Augmented	Phytophthora	942
Augmented	Scab	864
Augmented	Styler and Root	1063
TOTAL IMAGES		3183